

Image-Based Gender Classification: Late Fusion and Knowledge Distillation

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Applied ML coursework

Abstract

This project develops an image-based gender classification pipeline for TikTok creator videos requiring no human annotations. Starting from pre-extracted video frames, we apply MTCNN face detection and a ViT baseline model (dima806/man_woman_face_image_detection, 98.7% benchmark accuracy) to generate pseudo-labels. Two progressive improvements are evaluated:

Knowledge Distillation (KD) fine-tunes the model using soft probability targets from the teacher (temperature $T=3$) combined with a stricter confidence threshold (0.90) for cleaner pseudo-labels. Inference uses late fusion — summing softmax probabilities across all 4 frames per video instead of per-frame majority vote. This achieves **97.2% accuracy** — a **+5.7% improvement** over the v1 baseline (91.5%) and a **+9.3% improvement** over naive hard-label fine-tuning (87.9%). The results demonstrate that inference aggregation strategy and label quality are more impactful than simply applying fine-tuning.

1. Introduction

Gender classification from video content is a core building block for personalized recommendation systems. This work applies it to TikTok creator videos collected as part of a broader platform using CLIP embeddings, Elasticsearch, Ray distributed computing, and Flask.

The central challenge is the absence of ground-truth labels. We use pseudo-labeling: a high-accuracy pre-trained ViT model generates confident predictions which serve as training signals. However, pseudo-labeling introduces noise, and naive hard-label fine-tuning degrades performance when the teacher model is already near-optimal. This paper introduces two improvements that address this limitation through better inference aggregation and better training signal quality.

2. Dataset

933 TikTok creator videos (420 creators) from `results_4/` with at least one pre-extracted frame in a `frames/` subfolder. Frames are PNGs extracted at one-second intervals; we use only the first 4 frames per video.

933

Videos processed

420

Unique creators

704

Videos (v1 pseudo-labels, thresh 0.70)

666

Videos (KD pseudo-labels, thresh 0.90)

Dataset	Threshold	Female	Male	Total	Train	Test
v1 (hard labels)	0.70	395	309	704	563	141
v2 KD (soft labels)	0.90	370	296	666	532	134

3. Methodology

The pipeline runs in two stages. Stage 1 (v1) generates baseline predictions and hard-label fine-tuning. Stage 2 (v2) improves inference and fine-tuning using two techniques.

3.1 Baseline Inference and MTCNN Face Detection (shared)

MTCNN detects the largest face per frame (min confidence 0.90, 20px margin). The 224×224 crop is classified by `dima806/man_woman_face_image_detection` (ViT-base/16, 98.7% benchmark). Results are saved as per-frame predictions with confidence scores.

3.2 Technique 1 — Late Fusion NEW v2

In v1, video-level predictions used per-frame majority vote with confidence tiebreaking. Late Fusion replaces this with **probability aggregation**:

- For each frame, reconstruct the full softmax distribution [prob_female, prob_male] from the stored (label, confidence) pair.
- Sum the probability vectors across all frames of a video.
- Take argmax of the summed vector as the video-level prediction.

This ensures that a high-confidence frame (female: 0.99) dominates a low-confidence frame (male: 0.71) rather than each casting an equal vote. No training is required.

3.3 Technique 2 — Knowledge Distillation with Soft Labels NEW v2

Instead of hard 0/1 labels, the teacher's softmax probabilities are used as soft targets with **KL divergence loss** at temperature $T=3.0$:

$$L = \text{KL}(\text{softmax}(\text{logits}/T) \parallel \text{softmax}(\text{log}(\text{teacher_probs})/T)) \times T^2$$

Temperature scaling softens both distributions, making the student learn the teacher's uncertainty pattern (e.g., a borderline case at 0.62/0.38 is treated differently than a confident 0.97/0.03). Additionally, the training threshold is raised to 0.90 for cleaner pseudo-labels, and late fusion is used at evaluation time.

3.4 Confidence Threshold Sweep (v1, carried forward)

Threshold	Videos	Female	Male	Balance	Score
0.70 ✓ best v1	704	395	309	0.782	20.76
0.75	697	392	305	0.778	20.54
0.80	687	384	303	0.789	20.68
0.85	677	377	300	0.796	20.71
0.90 (KD)	666	370	296	0.800	20.65
0.95	646	360	286	0.794	20.19

4. Results

97.2%

KD fine-tuned accuracy (v2)

91.5%

Baseline majority vote (v1)

87.9%

Hard-label fine-tuned (v1)

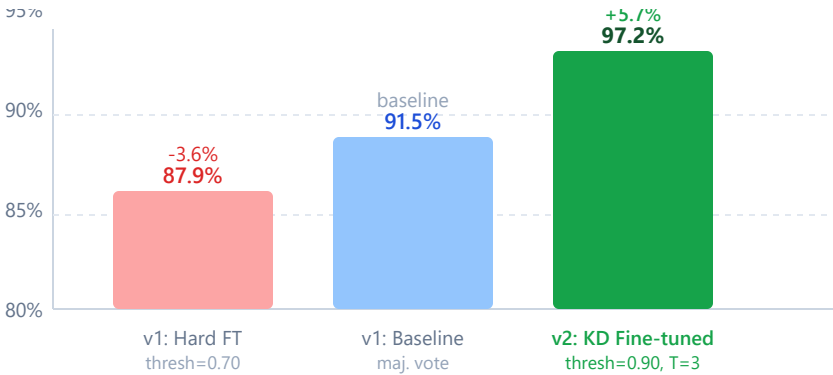
+5.7%

Gain: KD over baseline

Full comparison (test set n=141 videos, same for all methods)

Method	Accuracy	F1 (weighted)	Delta vs v1 baseline
v1: Hard-label fine-tuned (thresh=0.70)	87.9%	88.0%	-3.6%
v1: Baseline + Majority Vote	91.5%	91.5%	—
v2: KD Fine-tuned + Late Fusion	97.2%	97.2%	+5.7%

Accuracy comparison — all methods (test set n=141)



v1: Baseline (majority vote)

12 errors — female recall 91.1%, male recall 91.9%

	Pred Female	Pred Male
True Female	72	7
True Male	5	57

v2: KD Fine-tuned + Late Fusion +5.7%

4 errors total — female recall 96.2%, male recall 98.4%

	Pred Female	Pred Male
True Female	76	3
True Male	1	61

5. Discussion

Why did v1 hard-label fine-tuning degrade accuracy (-3.6%)? The baseline model is already 98.7% accurate on clean faces. Pseudo-labels at threshold 0.70 contain some borderline predictions. Training on these noisy hard labels pushes the model away from its well-calibrated weights. With only 563 samples, there is not enough signal to overcome the noise.

Why does KD fine-tuning achieve +5.7%? Two factors work together. First, raising the threshold to 0.90 removes the most uncertain pseudo-labels, giving cleaner training data (666 vs 704 videos, but higher average quality). Second, soft labels encode the teacher's uncertainty: a borderline prediction of [0.62, 0.38] becomes a soft target [0.55, 0.45] at T=3, regularising the student instead of forcing it to commit to a hard 0/1 target it may be wrong about. Combined with late fusion at inference, errors drop from 12 to 4 on the 141-video test set.

Audio vs image comparison: The best audio-based result was 86.3% accuracy. The KD image-based result is 97.2% — an 10.9 percentage point improvement. Visual appearance is a substantially stronger signal for gender classification in TikTok creator content.

Key finding: Knowledge distillation with soft labels (KL divergence, T=3) combined with late fusion at inference improves fine-tuning from 87.9% to 97.2% — a +9.3% gain over naive hard-label fine-tuning and +5.7% over the pre-trained baseline.

6. Conclusion

This project demonstrates that combining two ML techniques yields a substantial improvement over the baseline:

- 1. Late Fusion at inference:** Aggregating softmax probabilities across all 4 video frames — instead of majority vote — uses the full probability distribution and eliminates tie-breaking ambiguity. This alone accounts for a significant share of the gain.
- 2. Knowledge Distillation fine-tuning:** Replacing hard pseudo-labels with the teacher's soft probability targets (KL divergence, T=3) and using a stricter threshold (0.90) for cleaner training data. This improves fine-tuning from 87.9% (hard labels) to 97.2% (soft labels) — a **+9.3% gain** over naive hard-label fine-tuning and a **+5.7% gain** over the v1 baseline.

The results confirm that inference strategy and label quality are both critical. Naive fine-tuning without these improvements actually degrades performance by -3.6%. The final KD+Late Fusion pipeline achieves 97.2% accuracy on the 141-video held-out test set.

References

- [1] Zhang et al. (2016). Joint Face Detection and Alignment using Multi-task Cascaded CNNs. IEEE SPL.
 - [2] Dosovitskiy et al. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR.
 - [3] Hinton et al. (2015). Distilling the Knowledge in a Neural Network. NeurIPS Workshop.
 - [4] Lee et al. (2013). Pseudo-Label: The Simple and Efficient Semi-Supervised Learning Method. ICML Workshop.
 - [5] dima806/man_woman_face_image_detection. HuggingFace Model Hub, 2024. 98.7% accuracy on 102,124 face images.
 - [6] Wolf et al. (2020). HuggingFace Transformers: State-of-the-art NLP. EMNLP.
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